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An examination of secondary-level technology subject learning outcomes in Ireland

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Abstract
Secondary-level education in Ireland has recently completed one of the largest curricular reforms in the history of the state, resulting in the development of the Junior Cycle Framework for students typically aged 12-15. As part of this process, secondary-level technology subjects have undergone a major revision, shifting their positioning relative to each other and broader STEM initiatives. This paper employs a qualitative and quantitative content analysis approach to examine the learning outcomes specified for Applied Technology, Engineering, Graphics, and Wood Technology. Utilising the three-part Heuristic Framework for Technology Education, the learning outcomes of each subject were deductively and independently coded into categories of Technical Skills, Technological Scientific Knowledge, and Socio-ethical Technical Understanding. Discrepancies were resolved through consensus, and the frequency and distribution of these categories were calculated to support qualitative inference. The results reveal a predominant and distinct focus on Technical Skills across all four subjects, suggesting a curriculum heavily influenced by its vocational origins. Engineering and Graphics demonstrated the most significant skew; Engineering contained 23 Technical Skill outcomes out of 36, while Graphics was found to have no learning outcomes related to Socio-ethical Technical Understanding. While Applied Technology and Wood Technology demonstrated the highest proportional representation of Socio-ethical Technical Understanding, a considerable system-wide imbalance persists. These findings highlight a misalignment between stated subject aims and defined learning outcomes, informing critical recommendations for the ongoing reform of Senior Cycle technology subjects and future curriculum development.

Keywords Technology syllabus · Ttechnology education · Secondary-level education · Curricular reform

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Introduction

In Ireland, lower-secondary level technology education is offered through four subjects: Applied Technology, Engineering, Graphics, and Wood Technology. These subjects have recently undergone redevelopment as part of the most extensive curricular reform in the country's history and are now taught within the Junior Cycle Framework (JCF) (DES, 2015). The Junior Cycle is three years in duration, typically serving children between the ages of 12–15 (NCCA, 2009). The creation of the JCF involved a full review and redevelopment of all subjects. After completion of the Junior Cycle, most students continue on to complete the two-year Senior Cycle (upper-secondary level education). The Senior Cycle is currently in a similar redevelopment process that will see all subjects redesigned so that it integrates with the newly developed Junior Cycle and meets the changing needs of students as they prepare to leave secondary-level education.

The design process of the new Junior Cycle Framework, and specifications that define individual subjects, involved considerable input from pupils, parents, teachers, representative bodies, and school management. A major change in the new Junior Cycle is the shift to learning outcomes. Prior to this Junior Cycle reform, technology subjects had syllabi with subjects aims and objectives, but most of the intended learning was framed as explicit knowledge and skills that students should acquire. The formulation of the Irish technology curricula at that time mirrored the nature of the current Swedish curriculum for example, which describes the Technology subject intended learning in lists of "core content" (Skolverket, 2022). The current suite of technology subjects, and all Junior Cycle subjects, now focus on broader learning outcomes described by the National Council for Curriculum and Assessment (NCCA, 2019, p. 6) as "statements in curriculum specifications to describe the knowledge, understanding, skills and values students should be able to demonstrate after a period of learning." In this way, there has been a shift towards an outcomes-based philosophy, such that the formulation of the Irish technology subjects is now closer to that of countries such as South Africa (Ankiewicz, 2021), Malta (Pule, 2019), and England (Hardy, 2020), where these broader statements of learning are provided. "Learning outcomes provide the building blocks for teachers to plan their teaching, learning and assessment. Teachers can then use learning intentions and success criteria to take forward their planning and enable the learning outcomes to come alive in practice with their students." (NCCA, 2019, p.6). It is evident that learning outcomes are a fundamental force that heavily influence what is taught and how it is taught, and it is therefore essential that they are examined for multiple reasons.

One temporal reason relates to the ongoing reform of Senior Cycle education in Ireland, which includes a reform of technology education at this level. No subject area can be considered fully comprehensive, and this is especially true of technology subjects where there are several conceptions of what technology education should and could be (Dakers, 2024). Thus, the Junior Cycle technology offering will have specific curricular foci from any one subject's perspective. These unique points of emphasis differ depending on the perspective being taken; once understood, they can inform the coherence and progression of Senior Cycle technology education. Similarly, certain specialised domains may sit outside the primary scope of the classroom due to feasibility or because they are best realised beyond formal and/or general education systems. As an example, makerspaces often offer opportunity for technology learning in a way formal technology subjects cannot (Klapwijk, 2023).

In this spirit, there is value in identifying where informal education or outreach activity can complement and extend the offering of technology subjects. From this perspective, in this paper we offer a review of the current, newly reformed Junior Cycle technology learning outcomes in the Irish curriculum, not from a perspective of criticism, but of constructive critique with a view towards the continuous evolution of learning for all technology students. To give more context to this endeavour, the following sections provide a brief history of technology education in Ireland, followed by an illustration of the variation in formal technology education internationally and an argument for the importance of coherent local interpretations. Finally, we provide an overview of Nordlöf et al.'s (2022) heuristic framework for technology education, which is used to guide our curriculum analysis.

History of curriculum development in Ireland

The history of secondary-level curriculum development in Ireland has transitioned from a centralised, dual-track system to a modern, outcomes-based framework focused on economic and social integration. Historically, the Irish curriculum was characterised by state control and a reified view of knowledge as a fixed anthology of subjects (Gleeson, 2022). Curriculum development in Ireland has been shaped by a shifting balance between national identity, denominational influence, and economic utility. Following independence, a primary influence was the manner in which the state used the curriculum as a tool for national reconstruction and the restoration of the Irish language (Walsh, 2016). This era was also defined by a strong church-state partnership, where the Catholic Church exercised significant influence over educational philosophy and school management (Gleeson, 2022). A pivotal transition occurred in the 1960s with the OECD report, Investment in Education (1965), which shifted the system's institutional rationale from a personal development paradigm to a human capital paradigm focused on driving economic growth (O'Sullivan, 1992). In the modern era, development is heavily influenced by globalization and neoliberal market forces, which have introduced a focus on pre-determined learning outcomes, competences, and transferable skills (Gleeson, 2022).

Technology subjects in Ireland were initially offered within the vocational school system following the 1930 Vocational Educational Act. These vocational schools were created specifically for training and pre-employment preparation (Coolahan, 1981), and the subjects offered were responsive to "local needs", often industrial (Buckley, 2023). This type of school, distinct from existing secondary schools, created a dual system where traditional secondary schools had significantly higher status than their vocational counterparts. Consequently, vocational and technical education in Ireland traditionally held a lower social status within the secondary-level education system (NCCA, 2017). Until 1966 students could not progress to university from these vocational schools, further re-enforcing the class divide that was a distinct consequence of a two track secondary-level education system (Whelan & Hannan, 1999). Although changes in the late 1960s reduced differences between vocational and secondary school systems, traditional vocational subjects, now included in the secondary school system, retained their low status. While this undervaluation of vocational education is not unique to Ireland (Banks & Williams, 2022; Buckley, 2023), it has been recognised as a consistent force in the evolution of these subjects.

> "For generations, Vocational Education Committee (VEC) schools provided those who eventually became tradespersons with their basic education through the Group Certificate and, later on, the Intermediate/Junior Certificate. Critically, also, the VEC schools provided many who would take up apprenticeships and similar types of employment with their core vocational skills – woodwork, metalwork, mechanical drawing, home economics, typing and book-keeping." (ETBI, 2013).

Despite several revisions of the technology syllabi between original conception as technical education to current day technology education, this vocational focus has persisted.

> "The craft of the technology subjects, be they the bench skills in a practical room or the drawing skills in Technical Graphics, appear to be the main focus of learning across all the subjects." (NCCA, 2017, p.43).

In contrast, the vocational programmes that these subjects were initially designed to feed have now moved to a less process and skills orientated approach (ETBI, 2013). In response to a perceived heavy practical skill focus, which was no longer relevant to the vocational programs that had once been the raison d'être of technology education, a redevelopment of all technology subjects was undertaken as part of the Junior Cycle Framework. This process aimed to create "a modernised curriculum across all subjects" (DES, 2015, p.2). This resulted in the creation of the current suite of four technology subjects at Junior Cycle level, inclusive of the Applied Technology, Engineering, Graphics, and Wood Technology subjects, with reforms of Senior Cycle technology education currently underway.

International variations on technology education

In their discussion on the development of technology education, Jones et al. (2013) explore the emerging philosophical foundations, international curriculum developments, and the role of 'Technology' subjects within the broader STEM movement. They assert that the intended, implemented, and achieved curriculum varies not only by country but also by region, individual schools, and classrooms. Further, as can be seen in the edited collection by John R Dakers (2024), which includes 37 chapters on the prospective futures of technology education, there is significant variation in the views of potential technology education even amongst the academic community. Indeed, technology education can be quite varied, with international handbooks offering country level descriptions of curricular structure (Benson & Lunt, 2011; Jones & de Vries, 2009).

Jones et al. (2013) highlight the uncertain position of technology in many school curricula globally, as educational authorities strive to advance STEM initiatives alongside individual subjects. In regions where technology has been established as a standalone discipline, initiatives like STEM in the USA and UK, and VET in Australia, might lead to its integration into broader studies or its reclassification as a subset of other disciplines. Without a clear subject identity, such integration could eventually erode the distinct identity of this still-emerging field. That said, local interpretation significantly affects both teachers and students, leading to diverse implementations even though universal key messages can be applied. As can be seen by the issues faced by [Design and] Technology education in the UK (e.g. Barlex & Steeg, 2017; Spendlove, 2017), effective integration of technology into

general education requires a clear definition of the discipline, comprehensive teacher education and professional development, strong professional networks, well-defined educational pathways from early childhood to tertiary education, and a growing research culture focusing on graduate and postgraduate programs. A lack of relevance to specific professions has also been observed within an Irish context (Power & Buckley, 2024).

Banks and Williams (2022) explored recent developments, classroom experiences, and the potential future of technology education, drawing on examples from various countries including Australia, Bangladesh, Belgium, Brazil, China, Israel, Malawi, Taiwan, and the United States. They argue that the goal of technology education is to expand students' understanding of technology's societal impact, develop appropriate artifacts to meet specific needs, and tackle real-world problems. By applying sound reasoning, assessing facts, and using skills acquired through technology education, global challenges like the COVID-19 pandemic and climate change can be engaged with in greater depth by students and contribute to a more critical understanding. In technology curricula that are traditionally heavily vocationally focused, this balance will likely be more difficult to achieve (Power & Buckley, 2024).

Global variations in technology curricula and standards result in differing perceptions of what technology education is, should, or could be. This creates challenges in discussing and comparing technology education internationally. Additionally, discrepancies may exist at the national level or even within local schools, as teachers, principals, and researchers often hold diverse views on technology curricula and its teaching (Doyle et al., 2019). Unlike more established subjects such as mathematics and science, technology lacks a well-defined and coherent subject philosophy (Chesky & Wolfmeyer, 2015), compounding challenges related to examinations of technology curricula. However, this has allowed for considerable flexibility in adapting to regional societal, environmental and industrial contexts (Banks & Williams, 2022).

A framework for technology curricula examination

Nordlöf et al. (2022) propose a heuristic framework for technology education that builds upon established traditions of technological knowledge. They observe that most of the content in technology education is connected to the knowledge traditions of craftsmanship, engineering, the humanities and social sciences. By "knowledge tradition," they refer to the collective knowledge and skills that have evolved within a community that often loosely organises around similar tasks and shares tips, tricks, and information. These traditions develop organically, influenced by the needs and interests of the community members, and are shaped by both formal regulations and informal, constantly evolving norms. Unlike traditional epistemological categories defined by methods of justification, knowledge traditions are defined by the communities that practice them. These traditions are not strictly separate; their boundaries are often indistinct and overlapping (Håkanson, 2010). The proposed heuristic framework is structured as an epistemological tripod for technology education, supported by three interdependent pillars. Each pillar corresponds to a distinct category of technological knowledge, derived from the respective knowledge traditions mentioned previously: (1) Technical Skills, rooted in the craftsmanship tradition, (2) Technological Scientific Knowledge, stemming from the engineering tradition, and (3) Socio-ethical Technical Understanding, based on the humanities and social sciences tradition. The

first category of knowledge in the framework is technical skills. This type of knowledge originates from craftsmanship and other experience-based traditions, validated primarily through experience and trial and error (e.g., Ropohl, 1997; Vincenti, 1990). These skills are typically demonstrated through actions rather than verbal explanations. Technical skills are taught in craft and technical education programs and are utilized by craftsmen and technicians in their trades. Historically, this form of knowledge has been a significant component of technology, imparted from master to apprentice. The emphasis is on functionality rather than understanding the underlying reasons. Common activities in this category include making, sketching, drawing, measuring, merging, and repairing. Although the methods and tools of craftsmen have evolved over time, the primary focus on practical effectiveness has remained constant throughout history.

The second category is technological scientific knowledge. As the name suggests, this involves technological knowledge acquired through a general scientific approach, although not necessarily rooted in natural science. This type of knowledge predominantly stems from engineering and natural science traditions and is frequently represented using mathematical models (Hansson, 2013). It is often validated through scientific methods, although standards and practices (Norström, 2014) also serve as foundations. In this category, understanding why things work is often as crucial as knowing that they work. Typical activities could include analysing, calculating, describing, measuring, documenting, and engineering drawing. Engineering is primarily an academic knowledge tradition, learned mainly through formal education. While some engineering methods are based on experience, they are primarily applied through scientific experiments, methodical testing, or formal standards. Explanation and modelling play significant roles in the engineering tradition.

The third category is socio-ethical technical understanding, which addresses the relationship between technology and the human world. This is knowledge about technology rather than within it, viewed from the perspective of a critical philosophy of technology (de Vries, 2005; Vries, 2016). It focuses on human actions and diverges from the first two categories that emphasize the creation and use of technological artifacts or systems. Socio-ethical technical understanding highlights sociological, ethical, political, and environmental aspects of technology. This type of knowledge is developed among ethicists, historians, and sociologists of technology and includes insights into sociological, historical, political, and ethical perspectives on the role of technology in society. The primary goal is to teach students to discuss and understand various aspects of technology over time (past, present, and future) and the reciprocal interactions between technological development and societal evolution. This knowledge is justified by research methods from the social sciences and humanities. Typical activities could include describing, evaluating change over time, analysing, and evaluating from ethical and sustainable development perspectives.

Aim of the study

The current study aims to examine the balance of learning outcomes across the Irish Junior Cycle suite of technology subjects through evaluation and categorisation using the three-part heuristic framework for technology education developed by (Nordlöf et al., 2022). Similar content analyses of technology curricula has been conducted previously, such as by Lin and colleagues in Sweden (Lin et al. 2024a, b) from a perspective of educational

potential. As previously noted, this study adopts a similar lens of potential for augmentation, supplementation and an enhanced learning experience of technology learners.

Methodology

Research design

A mixed qualitative and quantitative content analysis approach is used in this study, enabling a thorough examination of the learning outcomes specified in the subject. This method is selected for its capacity to provide insights into the structure and content of the subject specifications (Forman & Damschroder, 2007). From an ethical perspective, the study utilised publicly available subject specifications which are national curriculum documents. Acknowledgment of sources and suitable attribution were maintained throughout the analysis.

Data collection

Learning outcomes contained within Junior Cycle technology subject specifications (Department of Education and Skills, 2018a, 2018b, 2018c, 2018d) were categorised. Each specification outlines the subject rationale and integration with the broader Framework for Junior Cycle (NCCA, 2024) which outlines the broader educational objective for all students. This Framework includes 24 Statements of Learning and associated key skills that schools must ensure are represented in all children's educational experience. The primary units of analysis were the learning outcomes listed in the subject specification documents, as these are central to the student learning experience. Additionally, considerations of claimed cross-curricular links and subject aims are also included.

Category definition

Categories were deductively defined based on the three-part heuristic framework for technology education created by Nordlöf et al. (2022). Nordlöf et al. (2022) note that "One of the main purposes of the framework is to provide a tool for the analysis and comparison of technology curricula. Even though technology education varies in different countries, its content can mainly be reduced to the three categories in our framework" (p. 1595). The three-part heuristic framework supported categorisation of each learning as either (1) Technical skills, (2) Technological Scientific Knowledge or (3) Socio-ethical Technical Understanding.

Coding and analysis

The subject specifications were thoroughly reviewed, and relevant content was deductively coded based on the three aforementioned predefined categories. A coding guide was developed to ensure consistency and reliability. To enhance reliability, both authors independently coded the curriculum documents. Discrepancies were resolved through discussion and consensus aligned with an interpretivist paradigm rather than considering a statistical measure of reliability. The frequency and distribution of each category within the subject

specification were calculated only to support qualitative inference as it is recognised that quantity of learning outcomes does not necessarily translate to enacted pedagogical time, or depth of associated learning. Qualitative analysis focussed on interpreting the significance and context of each category's presence.

Interpretation and reporting

The learning outcomes analysis was examined in conjunction with the stated subject aims and cross-curricular links. Patterns and trends were identified to gauge the broader implications for the subject's relevance to future technology-related study paths. Findings were documented with supporting examples from the subject specification. The analysis was presented to emphasise key insights and offer recommendations for future curriculum development. This structured approach provides valuable insights into the structure and focus of technology education curricula, potentially informing educators, policymakers, and curriculum developers in their efforts to enhance technology education. The following "Findings" section presents the analysis of each subject individually with a standardised comparison at the end, and this is followed by qualitative interpretation across subjects in the "Discussion" section.

Findings

Applied technology

Within the Applied Technology subject specification 26 Learning Outcomes are presented (Department of Education and Skills, 2018a). Fifteen of these outcomes were categorised as Technical Skills, 7 were categorised as Technological Scientific Knowledge and 4 as Socio-ethical Technical Understanding (Table 1).

As with all subjects considered within this evaluation, Applied Technology demonstrates a distinct focus on Technical Skills. However, this is not as extreme as the Engineering subject, and Applied Technology has the highest proportion of Socio-ethical Technical Understanding outcomes of all subjects (Fig. 1).

Engineering

The Engineering subject specification demonstrates a stark focus on Technical Skills, with 23 of the total 36 learning outcomes categorised as such. Twelve of the total 36 learning outcomes categorised as Technological Scientific Knowledge. Only 1 of the total 36 learning outcomes was categorised as Socio-ethical Technical Understanding (Table 2).

The scarcity of Socio-ethical Technical Understanding learning outcomes is seemingly inconsistent with the subject specification. Within the specification document the subject claims to support 2, of a total of 6 listed, of the broader Junior Cycle "Statements of Learning" (SOL) that directly relate to socio-ethical factors. SOL 19: The student values the role and contribution of science and technology to society, and their personal, social and global importance. SOL 24: The student uses technology and digital media tools to learn, work and think collaboratively and creatively in a responsible and ethical manner. (Department of

Table 1 Categorisation of applied technology learning outcomes

Technical Skills
1.3 refine ideas through the use of prototyping
1.4 review planning decisions throughout
3.1 analyse the impact of constraints on the design of solutions
1.7 apply innovative approaches in design solutions
2.3 recognise the principles of control systems when developing their solution
2.4 design a logical sequence of instructions to control a device or system
2.5 apply innovative approaches to designing control system solutions
1.9 select appropriate materials, equipment and processes in solving a problem
1.10 execute a plan using appropriate tools, materials and processes
1.11 demonstrate adherence to recognised health and safety standards
2.7 identify appropriate energy and control systems for design solutions
2.8 create control solutions to identified problems
1.12 document progression from concept to realisation
1.13 communicate evidence of the iterative process of design
2.9 communicate technical information in appropriate form
Technological Scientific Knowledge
1.1 develop a design solution drawing on experience and using evidence, reasoning, and decision making
1.2 analyse problems using a systematic approach
2.1 investigate relationships between the inputs, transformations, and outputs occurring within simple control systems
2.2 evaluate ideas through the use of simulation
3.2 evaluate the effectiveness of solutions
1.8 develop a plan for the realisation of a solution
2.6 explore energy conservation and efficiency
Socio-ethical Technical Understanding
1.5 consider the end-user experience at each stage of the design process
1.6 understand the role, impact and potential of existing and emerging technologies
3.3 explain how human, societal and environmental considerations affect solutions and outcomes
3.4 explore applications of technology in local contexts

Education and Skills, 2018b, p.6). This strong claimed link is not reflected in the subject's learning outcomes. This raises questions related to the reflection of learning outcomes on claimed links between the subject and broader Junior Cycle, as well as concerns related to balance within the subject itself (See Fig. 2).

Graphics

Graphics, similar to the previous Engineering subject, places a heavy focus on Technical Skills with 23 of the 37 total learning outcomes being categorised as such (See Table 3).

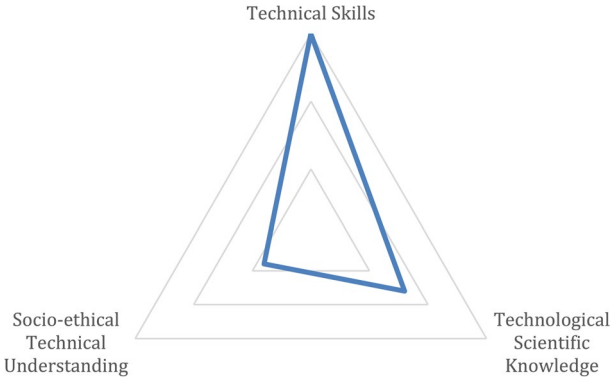


Fig. 1 Applied technology categorised learning outcomes

All of the remaining 14 learning outcomes were categorised as Technological Scientific Knowledge, with no learning outcomes being categorised as Socio-ethical Technical Understanding.

The omission of Socio-ethical Technical Understanding learning outcomes again presents an apparent mismatch between examples of learning presented within the specification document and the stated learning outcomes (Department of Education and Skills, 2018c). For example, the specification claims to serve Statement of Learning 19 "SOL 19 Values the role and contribution of science and technology to society, and their personal, social and global importance" and provides the following example of relevant learning "Students will evaluate the impact of technologies on their lives, society and the environment." (p.6). However, not a single Graphics learning outcome reviewed within the current study was identified as relevant (Fig. 3).

Wood technology

While Wood Technology maintains a distinct focus on Technical Skills, it is not as heavily focused as Engineering. Out of a total of 38 learning outcomes, 18 were categorised as Technical Skills, 14 as Technological Scientific Knowledge and 6 as Socio-ethical Technical Understanding (Table 4).

Wood Technology demonstrated the highest proportional representation of learning outcomes categorised as Technological Scientific Knowledge and Socio-ethical technical Understanding learning outcomes (See Fig. 4).

Comparisons across subjects

Noting that the total number of learning outcomes per subject specification varies considerably, overall weighting was calculated based on percentage for each category relative to the subject total. When considered as a totality, a considerable focus on Technical Skills is evident (See Fig. 5). Applied Technology and Wood Technology shows similar profiles, while Engineering and Graphics demonstrate an extreme skew with only a single learning outcome being categorised as Socio-ethical Technical Understanding in Engineering and none in Graphics.

Table 2 Categorisation of engineering learning outcomes

Technical Skills
1.1 understand the concepts and approaches that are required when solving an engineering problem
1.2 demonstrate a range of manufacturing processes
1.3 recognise and adhere to health and safety standards
2.1 understand the key stages of the engineering design process
2.2 evaluate the factors that influence design
2.3 choose a suitable material to engineer a product
3.1 explain the operation of basic mechatronic systems
3.2 investigate relationships between inputs, processes and outputs for basic control systems
1.6 engage with the various engineering disciplines by relating them to everyday applications
3.4 explore the application of systems in an engineering setting such as the classroom, home and industry
3.6 configure and program basic mechatronic systems using appropriate software
3.7 design a basic mechatronic system either individually or collaboratively
1.8 identify appropriate tools and equipment specific to a task
1.9 apply suitable manufacturing processes to engineer a product
1.10 demonstrate high-quality work, to include accuracy and surface finish
2.8 manufacture a product from a working drawing
2.9 modify an existing product/design
2.10 incorporate basic project management techniques
1.11 create sketches, models and working drawings
1.12 interpret working drawings
1.13 use appropriate technical language and notations
2.11 present ideas through modelling and prototyping, using appropriate media
3.10 represent key information using appropriate media
Technological Scientific Knowledge
1.4 understand the properties associated with a range of engineered materials
3.3 appreciate the application of mechanisms in a controlled system
1.5 research applications of existing and emerging technological developments
2.4 explore how design impacts on the function and quality of a product including ergonomic considerations
2.5 apply appropriate engineering concepts and approaches in the execution of their design solutions
2.6 use relevant information to enhance design and function
1.7 develop engineered solutions to various challenges
2.7 apply their knowledge of the properties associated with a range of engineering materials
3.8 build and test a basic mechatronic system with specific inputs or outputs
3.9 incorporate basic mechatronics into their engineered products
2.12 communicate their design decisions using suitable media
3.11 justify their choice of the most appropriate system or systems for a specified purpose
Socio-ethical Technical Understanding
3.5 investigate the impact of mechatronics on the environment and society

The distribution of learning outcomes highlights a continued focus on processes and manufacturing, despite previous calls to broaden the learning experience provided within technology subjects in Ireland (NCCA, 2017).

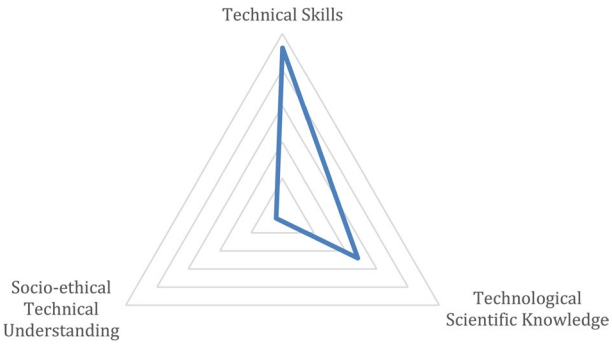


Fig. 2 Engineering categorised learning outcomes

Qualitative interpretation and discussion

Qualitative interpretation and systhesis

The application of the tri-heuristic framework developed by Nordlöf et al. (2022) to evaluate learning outcomes in Applied Technology, Engineering, Graphics, and Wood Technology identifies a significant opportunity for the strategic evolution, augmentation, and supplementation of these curricula. While the current emphasis on Technical Skills reflects a strong foundation in practical competency, the observed distribution across Technological Scientific Knowledge and Socio-ethical Technical Understanding provides a clear opening for future pedagogical offering. The existing homogeneity across the four subjects suggests a fertile ground for curriculum designers to intentionally diversify epistemic structures, moving beyond a primary focus on technical proficiency to incorporate a broader range of knowledge traditions, particularly into the domain of socio-ethical technical understanding. By addressing the identified inconsistencies between stated subject aims and explicit learning outcomes, there is a transformative potential to achieve greater alignment within the specifications. Although achieving such curricular balance is recognised as a complex task in vocationally oriented education (Power & Buckley, 2024), these findings highlight a pathway for the robust integration of socio-ethical technical understanding, ultimately fostering a more holistic and modernised model of technology education (de Vries, 2005; Vries, 2016).

Within this analysis it is clear that a distinct focus on Technical Skills is evident across all subjects, despite prior reviews calling for a more balanced approach to technology education in Ireland (NCCA, 2017). However, as noted previously it is worth reiterating that the quantity of learning outcomes categorised within one epistemic category does not reflect time and depth of learning absolutely. Further qualification from insight into enacted practice is warranted to better understand the translation of the written to the enacted curriculum. That said, there does appear to be a heavy focus on Technical Skills evident in all technology subjects evaluated, and it is likely that this is leading to a lower-secondary level technological education within Ireland that is heavily focused on technical skill development in the translation of these specifications to enacted practice.

Research employing the tri-heuristic framework indicates that socio-ethical technical understanding, defined as knowledge about technology encompassing sociological, ethical, political and environmental perspectives, is consistently the most neglected pillar in tech-

Table 3 Categorisation of graphics learning outcomes

Technical Skills
1.1 visualise the manipulation of 2D shapes
1.2 analyse graphical information for the planning of a 2D solution
1.3 derive 2D solutions using appropriate media
2.1 visualise the manipulation of 3D objects
1.5 illustrate ideas using free-hand sketches to accurately communicate their thought process
1.7 interpret and create graphical representations of data/information
2.4 appreciate the role of 3D graphics in the creation of solutions
2.6 apply their understanding of 3D principles to solve problems
2.7 construct solutions to presented and/or defined problems
1.8 communicate the progression of ideas and thinking during the course of an activity using a variety of media
1.9 represent 3D information using 2D conventions
2.8 construct a 3D representation of an artefact or abstract idea using a variety of media and methods.
2.9 communicate the progression of ideas/thinking during the course of an activity using a variety of media
3.6 develop design ideas/solutions through modelling and prototyping using a variety of media
3.7 use computer-aided graphics to communicate design solutions effectively
3.8 represent graphically their approach to a design task
3.9 apply a variety of rendering and presentation techniques to enhance the communication of solutions
1.10 understand the properties of geometric shapes
1.11 appreciate the application of geometric constructions in the study of other areas
1.12 construct 2D solutions accurately in accordance with graphical conventions.
2.10 understand the properties of geometric objects and surfaces
2.11 appreciate the application of geometric principles in the study of other areas
2.13 apply geometric principles to construct accurate 3D solutions in accordance with graphical conventions.
Technological Scientific Knowledge
2.2 analyse graphical information for the planning of a 3D solution
2.3 derive 3D solutions using appropriate media
3.1 recognise 2D and 3D features in everyday objects and artefacts
3.2 appreciate the hidden features of an object or an artefact necessary for its representation
3.3 demonstrate their spatial understanding by modelling and/or simulation
1.4 appreciate the role of 2D graphics in the creation of solutions
1.6 apply their understanding of geometric principles to solve problems
2.5 develop ideas using free-hand sketches and other media to accurately communicate the thought process
3.4 solve real-context and abstract problems using graphical techniques
3.5 analyse and evaluate both their own work, and the work of others
2.12 generate and develop design ideas using appropriate geometric principles and constructions
3.10 investigate and apply the principles of plane and descriptive geometries to create solutions
3.11 investigate how geometric principles and constructions found in the natural world have provided inspiration for human applications
3.12 develop an appropriate graphical representation of a solution to a contextual problem of their choice

nology education curricula (Nordlöf et al., 2022). Investigations into various technology-based curricula have identified a significant systemic imbalance where a predominant focus on technical skills persists. Furthermore, empirical studies suggest that socio-ethical knowledge rarely emerges naturally in student discourse during design projects without specific

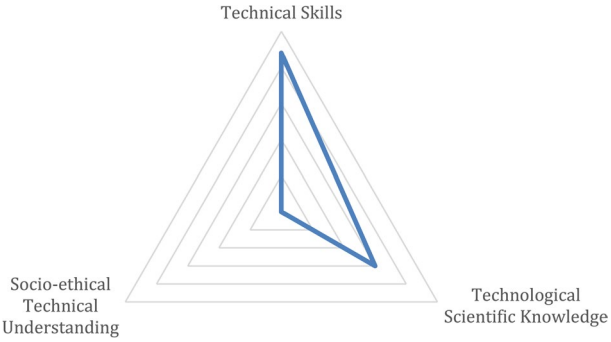


Fig. 3 Graphics categorised learning outcomes

instructional scaffolding (Citrohn, 2023). This difficulty in achieving curricular balance is often attributed to the enduring influence of vocational and craft origins, which prioritise functional know-how over the critical socio-ethical insights required for a comprehensive technology education (Power & Buckley, 2024). This lower level of affordance for socio-ethical technical understanding relative to technical skills and technological scientific knowledge emphasises the need for this type of learning to be made explicit is curricular documentation, as while technical skill development may naturally emerge, socio-ethical learning is less likely, and in modern society this opportunity for learning, we contend, is paramount.

Building upon these insights, curriculum developers, and those involved in educational provision in the variety of formats it can take, are presented with the opportunity to enhance the Irish technology education experience by ensuring the current Junior Cycle offering is balanced through, for example, the new Senior Cycle subjects, the incoming primary level reformed curriculum, and through informal and non-formal educational opportunities. By intentionally aligning the advanced specifications of the Senior Cycle with a more equitable distribution of the tri-heuristic pillars, curriculum designers can create a progressive educational trajectory that transition students from foundational technical competency toward a more sophisticated mastery of technological scientific knowledge and socio-ethical technical understanding. Furthermore, the incoming inclusion of technology education at the primary level offers a fertile ground for introducing these diverse knowledge traditions early. In Ireland, technology learning is being introduced into primary level education from the 2026/27 academic year, enacted through two strands and learning called Technology and Engineering within an integrated STEM framework. This will mark the introduction of compulsory technology learning for all Irish students as secondary-level technology subjects are optional and brings with it opportunity to provide an early foundation into technical skills development, technological scientific knowledge, and socio-ethical technical understanding. Such a cohesive, longitudinal approach to formal technology education provision would empower students to engage with technology as a multifaceted discipline, ensuring they possess the critical and analytical tools necessary to navigate the complex sociological and environmental challenges of the modern world. However, it is critical that learning outcomes are aligned with the overall espoused values and purpose of the subjects.

In some instances, there is a lack of alignment between the stated aims of the subjects considered within the current analysis and their listed learning outcomes. Taking the Engineering subject specification as an example, which claims to support the broader Junior Cycle curriculum through a third of its included “Statements of Learning” directly relating

Table 4 Categorisation of wood technology learning outcomes

Technical Skills
1.1 explore key elements required for the completion of tasks
1.2 justify the selection of plans, processes and materials for the completion of tasks
1.3 collaborate effectively in a workshop learning environment
1.4 manage themselves and their resources
2.1 explore design problems
2.3 evaluate their own progress to inform future learning
3.1 identify common species of trees
1.5 represent key information graphically
1.6 create sketches and working drawings to recognised standards using a variety of media
1.7 explain the function and application of a range of tools, equipment, fixtures and fittings
2.5 communicate relevant information
2.6 produce sketches, drawings and models/prototypes to explore design ideas
2.7 communicate a suitable approach to solving a problem
2.8 compile a folio through appropriate media
1.8 apply knowledge of and skills in a range of appropriate existing and emerging principles, processes and techniques
1.10 apply recognised health and safety practices in the use of tools, equipment and materials
2.10 devise templates and models using various media
2.11 produce purposeful, functional, appealing artefacts
Technological Scientific Knowledge
2.2 manage information and thinking to support an iterative design process
2.4 understand key principles of design and ergonomics
3.2 evaluate the characteristics and properties of common species of trees
3.3 understand the properties associated with a range of materials applicable to Wood Technology
3.4 evaluate the use of wood in comparison to alternative materials
3.5 explain the properties associated with the classification of wood
3.6 discuss the use of wood in comparison to alternative materials
3.7 justify the use of materials based on characteristics and properties within a context
1.9 demonstrate principles of craft excellence through the design and realisation of tasks and artefacts
2.9 evolve their solutions based on critical reflection
2.12 create an artefact having considered factors such as materials, cost, time resources and skills
3.8 utilise the natural aesthetics and properties of wood to enhance the appearance and function of an artefact
3.9 create an artefact that demonstrates an understanding of the properties associated with a range of materials applicable to Wood Technology
3.11 investigate the journey of wood from forest to end use
Socio-ethical Technical Understanding
3.10 appreciate the role of forestation and wood in terms of local/global ecology and sustainability
3.12 consider the impact on the natural environment when sourcing materials
1.11 investigate the environmental impacts of using wood as a natural and renewable resource
1.12 appreciate sustainable practice throughout their learning
2.13 recognise the environmental and social impacts of design decisions
2.14 investigate how to minimise material use and manage waste

to socio-ethical factors (Department of Education and Skills, 2018b, p.6), only a very small amount of the subject's learning outcomes can be categorised as Socio-ethical Technical Understanding. However, this content analysis alone cannot be used to comment on the

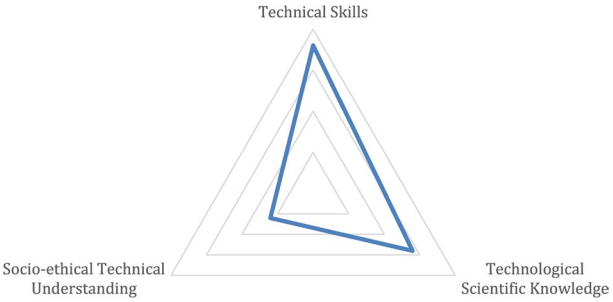


Fig. 4 Wood technology categorised learning outcomes

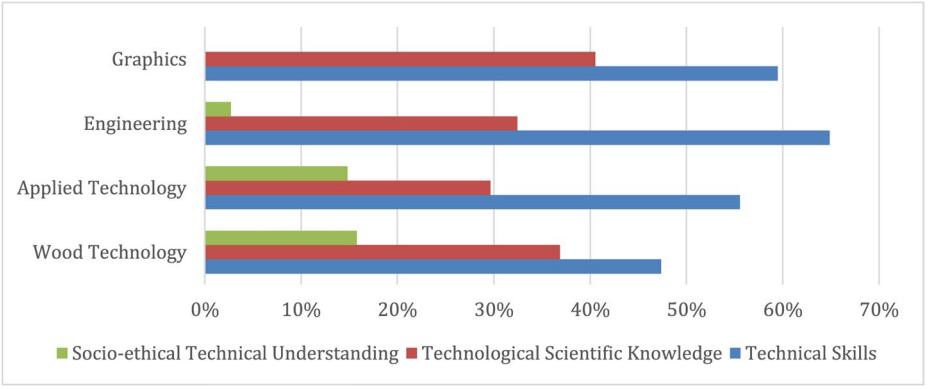


Fig. 5 Comparisons across subjects

translation of the specifications into practice. It is useful to triangulate this analysis with assessment practices as, as Eisner (2004) notes, assessment tends to define what is valued within a subject.

Assessment is a particularly contentious factor in Ireland's secondary-level education system (McGarr et al., 2022, 2023). For example, the Senior Cycle State Examinations which form the basis of university matriculation systems are heavily featured in national press each year (McCormack et al., 2020). It also represents a major point of contention in the current, and previous, reform efforts (MacPhail et al., 2018; Murchan, 2018), particularly with respect to the role of the teacher. In the context of this study, it should be noted that where the curriculum is governed and developed by one state body, the NCCA, national summative assessment in Irish secondary-level education is designed and overseen by another, the State Examinations Commission (SEC). Assessment at Junior Cycle involves two formative classroom-based assessments (CBA's) described by the NCCA and implemented and assessed by the student's regular classroom teacher, and the SEC organises the summative, state-certified examinations independently. As such, the NCCA develops the curriculum, and the SEC then has a responsibility to review this and determine valid means of assessing learning where the student's regular teacher is not directly involved. This is a contrast to countries such as Sweden where it is the responsibility of the student's teacher to design the assessment mechanism for learning at local classroom levels, and means that, in

Ireland, the primary concern of teachers is on translating the learning outcomes into pedagogical approaches rather that into assessment mechanisms.

Taking the Engineering subject as an exemplar, it is possible to reflect on the implications of synergies between learning outcomes and assessment. In the Engineering subject 2025 assessment guidelines (SEC, 2025), 70% of marks are issued based on the production of an artefact using a fully defined set of working drawings and standardised components. There is no current provision for a design element or sustainability considerations within this assessment. This would place 70% of assessment on one manufactured artefact that directly aligns with the Technical Skill categorisation, further supporting the heavy focus identified in the review of the learning outcomes (See Fig. 4). This has the potential to impact student learning performance in the long term (Buckley et al., 2018).

The implications of this from the perspective of the current analysis are two-fold. First, it would seem that the assessment mechanism does reflect a subject which emphasises Technical Skills over Technological Scientific Knowledge and Socio-ethical Technical Understanding. However, as argued by Lange and Meaney (2012), assessment heavily impacts how and what is taught. Thus, while through our content analysis of learning outcomes we are hesitant to conclude that the weighting of learning outcomes on Technical Skills would translate into practice, triangulating this with the assessment certainly suggests this to be the case. Second, as is argued by Brown (2004) and Hopmann (2015), assessment can narrow the curriculum as ultimately a measure becomes a target, and without suitably sophisticated assessment practices it will negatively impact the learning experience of students. While Technical Skills are emphasised in the subject assessment, the assessment is not broadly representative of the entire suite of Technical Skills learning outcomes presented in the specifications. The assessment focuses heavily on a subset of manual processing activities within the Technical Skills category.

Based on the analysis presented within this paper there appears to be some disconnect between learning outcomes, specification rationales and assessment. The use of a Tri-Heuristic Framework provides a novel perspective to highlight imbalances. Specifically, there appears to be a strong emphasis in Junior Cycle technology education on Technical Skills at the expense of Technological Scientific Knowledge and Socio-ethical Technical Understanding. There is always room for this to change in terms of enacted practice, and over time the assessment mechanism may change and consequently translate to a different weighting in practice, or there may be natural shifts influenced by, for example, changes in societal relationships with technologies and/or teacher professional development. What seems clear at present is that if efforts are to be invested in augmenting the student learning experience – whether through other formal provision, outreach initiatives, informal or non-formal learning activities, or otherwise, it is likely that Socio-ethical Technical Understanding is an area which is relatively underserved in the current offering, followed by Technological Scientific Knowledge. In the same vein, given the ongoing reform of Senior Cycle Technology Education, it is possible that the view is that Junior Cycle would focus on the provision of an underlying technical skillset to afford potential for higher level socio-ethical and scientific learning in upper-secondary level. Assuming the current heavy focus is an intentional position, then a heavier focus on Technological Scientific Knowledge and Socio-ethical Technical Understanding in the Senior Cycle Technology subjects could provide a more holistic learning experience when considering secondary-level education as a whole.

Enhancing socio-ethical technical understanding

The qualitative analysis of the Irish Junior Cycle technology suite reveals a significant epistemological imbalance, with Technical Skills learning outcomes averaging over 55 per cent across the four subjects. This pronounced skew is most evident in Graphics, which contains no socio-ethical outcomes, and Engineering, where such factors account for only 2.8 per cent of the written curriculum. This focus reflects what de Vries (2005) identifies as an overemphasis on descriptive knowledge – knowing how a system functions, while neglecting the normative component that defines technological knowledge. Unlike scientific knowledge that seeks truth, technological knowledge is inherently concerned with effectiveness and value judgements. By privileging procedural mastery, the curriculum promotes an instrumental view of technology as a neutral tool, ignoring the reality that technical artefacts are intrinsically value laden (de Vries, 2005).

If this imbalance is to be addressed, educators must incorporate the dimension of volition, which de Vries (2016) describes as the human will and values driving technological development. The current findings suggest that students are primarily taught about technical actions and objects, yet they are rarely prompted to engage with the motives behind these creations. Without learning outcomes that require students to ask what kind of world they should be building, technology education fails to achieve its transformative potential. de Vries (2016) argues for the epistemic emancipation of the subject, moving away from an applied science paradigm toward a holistic understanding that integrates the social sciences. This integration is essential for fostering technological literacy that encompasses not just functional capability but also the ability to make ethical judgements within a democratic society.

This curricular imbalance is further problematised by John R Dakers (2022), who critiques the persistence of 19th century craft based models in a 21st century world facing the existential threat of the Anthropocene. John R Dakers (2022) argues that a pedagogy of transmission, where teachers hand down pre-given technical content, prevents students from developing ethnotechnological literacy. This form of literacy requires young people to perceive technology as a fundamentally human activity situated in a sociocultural context rather than as a series of static, industrial facts. The high weighting of technical skills in the Irish curriculum reinforces a state of repetition. Learning what existing technologies do, rather than a state of becoming, where students are empowered to reimagine and transform their technologically textured lifeworld (John R Dakers, 2022).

Addressing the apparent lack of socio-ethical technical understanding lends to the implementation of a nomadic pedagogy, which prioritises flexibility and the formation of a community of inquiry. As John R Dakers (2022) suggests, this approach shifts the focus from the machine back to mankind by encouraging students to reflect on their own situation and values. Alternatively, a history-based curriculum would support similar pedagogical goals, where learners reflect on the development and societal impact of previously invented technologies as a means to consider the future (John R. Dakers & Hallström, 2025). Practical implementation could involve debating the value architecture of emerging technologies, such as Artificial Intelligence, or in considering the environmental impacts of the very artefacts they are creating. John R Dakers (2024) envisions a future for technology education where students move from being mere consumers of technical skills to active participants in sustainable development. By integrating these socio-ethical connections, technology sub-

jects can provide a more balanced epistemological tripod, preparing students to engage with global challenges like climate change through a critical and creative lens (John R Dakers, 2024).

Conclusion

While the current study focuses on the Irish context, it does carry implications for the global technology education community, as many nations face similar challenges while redeveloping their curricula to align with modern STEM initiatives (Jones et al., 2013). Specifically, the tension between preserving craft-based traditions and adopting design-driven frameworks is a shared phenomenon, with jurisdictions like the United Kingdom demonstrating a comparable focus on manufacturing processes at the expense of broader socio-ethical literacy (Power & Buckley, 2024). This research highlights the utility of the three-part heuristic framework (Technical Skills, Technological Scientific Knowledge, and Socio-ethical Technical Understanding) as a diagnostic tool for international policymakers to identify and address curricular imbalances (Nordlöf et al., 2022).

This study also highlights the critical role of learning outcomes in shaping what is taught, how it is taught, and how it is assessed within the context of the Irish technology education curriculum. The analysis reveals a pronounced emphasis on Technical Skills across all subjects. This focus appears to be a remnant of the historically vocationally oriented subjects, despite calls for a more balanced approach to technology education in Ireland. Assessment practices further reinforce the heavy focus on technical skills and this narrow assessment focus likely constrains the subjects from reaching their stated aims and limits students' learning experiences. The contentious nature of assessment in Ireland's secondary education system underscores the importance of thoughtful reform (McGarr et al., 2023, 2024).

In conclusion, to enhance the relevance and effectiveness of technology education in Ireland, it is essential to address the identified misalignments between learning outcomes, assessment practices, and stated educational aims. A more balanced and inclusive approach, incorporating socio-ethical considerations, design, and scientific elements could better prepare students for future educational pathways that are associated with the technology disciplines explored here. These areas present space for consideration for those providing informal and non-formal technology education initiatives and those involved in the reforms and implementation of a reformed Senior Cycle and new primary level curriculum. This study thus provides a foundation for further research and dialogue amongst educators, policymakers, and curriculum developers to foster a more comprehensive and holistic Irish technology education curriculum.

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